4. Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent, real-valued random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

a) Suppose all X_n, $n \in \mathbb{N}$ are uniformly distributed on $\{0, 1, \ldots, 9\}$. Show that $Y_n := \sum_{i=1}^{n} X_i 10^{-i}$, $n \in \mathbb{N}$ converges in distribution to a random variable Y uniformly distributed on $[0, 1]$. Does the sequence $(Y_n)_{n\in\mathbb{N}}$ also converge $\mathbb{P}$-almost surely to a uniformly distributed random variable on $[0, 1]$?

b) Suppose all X_n, $n \in \mathbb{N}$ are Cauchy$(0, 1)$-distributed and

$$M_n := \max\{X_1, X_2, \ldots, X_n\}, \quad n \in \mathbb{N}.$$

Show that $\frac{\pi M_n}{n}$ converges in distribution to a random variable M with distribution function $F_M(x) = \exp(-1/x)\mathbb{I}_{(0,\infty)}(x)$, $x \in \mathbb{R}$.

c) Suppose all X_n, $n \in \mathbb{N}$ are normally distributed, i.e., $X_n \sim \mathcal{N}(\mu_n, \sigma_n^2)$, $n \in \mathbb{N}$, and it holds that $X_n \xrightarrow{d} X$. Show that the distribution of X is then also a (possibly degenerate) normal distribution.

(2+2+2 Points)

Problem 4

Part a)

1. Convergence in Distribution: By definition, $X_k \sim \mathcal{U}(\{0, 1, \ldots, 9\})$, so its probability mass function is $\mathbb{P}(X_k = j) = \frac{1}{10}$ for $j \in \{0, \ldots, 9\}$. The characteristic function of X_k is:

$$\begin{aligned}
\varphi_{X_k}(u) &= \mathbb{E}[e^{iuX_k}] \\
&= \sum_{j=0}^{9} \mathbb{P}(X_k = j) e^{iuj} \\
&= \sum_{j=0}^{9} \frac{1}{10} e^{iuj} \\
&= \frac{1}{10} \sum_{j=0}^{9} e^{iuj} \quad \text{(geometric sum formula)} \\
&= \frac{1}{10} \frac{1 - e^{iu10}}{1 - e^{iu}}, \quad \text{for } u \neq 0 \qquad (1)
\end{aligned}$$

Because the random variables $X_1, \dots, X_n$ are independent, we have for $Y_n = \sum_{i=1}^n X_i 10^{-i}$:

$$\begin{aligned}
\varphi_{Y_n}(u) &= \prod_{i=1}^n \varphi_{10^{-i}\cdot X_i}(u) && \text{(by extending Lemma 3.5.6 via induction)} \\
&= \prod_{i=1}^n \varphi_{X_i}(u10^{-i}) && \text{(Theorem 3.5.2 b))} \\
&= \prod_{i=1}^n \frac{1}{10} \frac{1 - e^{iu10^{-i}\cdot 10}}{1 - e^{iu10^{-i}}} && \text{(by (1))} \\
&= \frac{1}{10^n} \prod_{i=1}^n \frac{1 - e^{iu10^{-i+1}}}{1 - e^{iu10^{-i}}} \\
&= \frac{1}{10^n} \frac{1 - e^{iu10^0}}{1 - e^{iu10^{-n}}} && \text{(telescoping product)} \\
&= \frac{1 - e^{iu}}{\frac{1 - e^{iu10^{-n}}}{10^{-n}}}
\end{aligned}$$

Let $x = 10^{-n}$. As $n \to \infty$, we have $x \to 0$. Then, for the denominator, we have:

$$\begin{aligned}
\lim_{n\to\infty} 10^n(1 - e^{iu10^{-n}}) &= \lim_{x\to 0} \frac{1 - e^{iux}}{x} \\
&\overset{\text{L'H}}{=} \lim_{x\to 0} \frac{-iue^{iux}}{1} \\
&= -iu.
\end{aligned}$$

Therefore, for any $u \neq 0$, the characteristic function converges to:

$$\lim_{n\to\infty} \varphi_{Y_n}(u) = \frac{1 - e^{iu}}{-iu} = \frac{e^{iu} - 1}{iu}.$$

For $u = 0$, $\varphi_{Y_n}(0) = 1 \to 1$, which continuously matches $\lim_{u\to 0} \frac{e^{iu}-1}{iu} = 1$. We have shown that for every $u \in \mathbb{R}$, the sequence of characteristic functions converges pointwise to a limit function $f(u)$:

$$\varphi_{Y_n}(u) \xrightarrow{n\to\infty} f(u) := \begin{cases} \frac{e^{iu}-1}{iu} & \text{if } u \neq 0, \\ 1 & \text{if } u = 0. \end{cases}$$

Let $Y \sim \mathcal{U}([0,1])$. Its characteristic function is:

$$\varphi_Y(u) = \int_0^1 e^{iux} \cdot 1\, dx = \left[\frac{e^{iux}}{iu}\right]_0^1 = \frac{e^{iu} - 1}{iu}.$$

Then $\varphi_{Y_n}(u) \xrightarrow{n\to\infty} \varphi_Y(u)$ for all $u \in \mathbb{R}$, and $\varphi_Y(u)$ is continuous at $u = 0$. By Lévy's Continuity Theorem (Theorem 4.2.4), $Y_n \xrightarrow{d} Y$, and, by the uniqueness theorem (Theorem 3.5.4), the limit distribution is $\mathcal{U}([0,1])$.

□

2. Almost Sure Convergence:

Yes, the sequence converges $\mathbb{P}$-almost surely. Since $X_k(\omega) \geq 0$ for all $k \in \mathbb{N}$ and all $\omega \in \Omega$, the sequence of partial sums $Y_n(\omega) = \sum_{k=1}^n X_k(\omega)10^{-k}$ is monotonically increasing for every $\omega \in \Omega$.

Also, the sequence is deterministically bounded from above, because the maximum value any X_k can take is 9:

$$Y_n(\omega) \le \sum_{k=1}^{n} 9 \cdot 10^{-k} = 9 \cdot \frac{10^{-1} - 10^{-(n+1)}}{1 - 10^{-1}} = 1 - 10^{-n} < 1, \quad \forall n \in \mathbb{N}.$$

Since $(Y_n(\omega))_{n\in\mathbb{N}}$ is a bounded, monotonically increasing sequence of real numbers, it must converge pointwise for $\omega \in \Omega$ (and, hence, $\mathbb{P}$-almost sure convergence, Definition 3.3.10). Let $Y^*(\omega) = \lim_{n\to\infty} Y_n(\omega)$, meaning $Y_n \xrightarrow{a.s.} Y^*$. Then, by Theorem 4.1.2 a), $Y_n \xrightarrow{p} Y^*$, and, by Theorem 4.1.8 a), $Y_n \xrightarrow{d} Y^*$.

From the first part of the problem, we already established that $Y_n \xrightarrow{d} Y$, where $Y \sim \mathcal{U}([0,1])$. Since both $Y_n \xrightarrow{d} Y^*$ and $Y_n \xrightarrow{d} Y$, then $\varphi_{Y^*}(u) = \varphi_Y(u)$. By the uniqueness theorem (Theorem 3.5.4), Y^* and Y share the exact same distribution, meaning $Y^* \sim \mathcal{U}([0,1])$.

Part b)

Step 1. Distribution Function of $\frac{\pi M_n}{n}$ By definition, $X_i \sim \text{Cauchy}(0,1)$, so its density is $f_X(t) = \frac{1}{\pi(1+t^2)}$. The corresponding distribution function is:

$$F_X(t) = \int_{-\infty}^{t} \frac{1}{\pi(1+s^2)}\, ds = \frac{1}{\pi}\arctan(t) + \frac{1}{2}.$$

Let $M_n = \max\{X_1, \ldots, X_n\}$ and $Y_n := \frac{\pi M_n}{n}$. For this scaled random variable sequence, we define the distribution function shorthand $F_n(x) := F_{Y_n}(x)$:

$$\begin{aligned}
F_n(x) &:= \mathbb{P}\left(\frac{\pi M_n}{n} \le x\right) \\
&= \mathbb{P}\left(M_n \le \frac{nx}{\pi}\right) \\
&= \mathbb{P}\left(X_1 \le \frac{nx}{\pi}, X_2 \le \frac{nx}{\pi}, \ldots, X_n \le \frac{nx}{\pi}\right) && \text{(definition of maximum)} \\
&= \prod_{i=1}^{n} \mathbb{P}\left(X_i \le \frac{nx}{\pi}\right) && \text{(by independence)} \\
&= \left[F_X\left(\frac{nx}{\pi}\right)\right]^n. && (2)
\end{aligned}$$

Step 2. Pointwise Convergence of the CDF:
We must evaluate the limit of $F_n(x)$ as $n \to \infty$ for all $x \in \mathbb{R}$ to apply Theorem 4.1.6. Because the necessary trigonometric identities require positive arguments, we partition the real number line into two cases:

1. *Case 1: $x \le 0$.*
   If $x \le 0$, then $\frac{nx}{\pi} \le 0$, which implies $\arctan\left(\frac{nx}{\pi}\right) \le 0$. Hence, $F_X\left(\frac{nx}{\pi}\right) \le \frac{1}{2}$.

$$0 \le F_n(x) = \left[F_X\left(\frac{nx}{\pi}\right)\right]^n \le \left(\frac{1}{2}\right)^n \xrightarrow{n\to\infty} 0.$$

2. *Case 2: $x > 0$.*
   For $t > 0$, we utilize the trigonometric identity $\arctan(t) + \arctan(1/t) = \frac{\pi}{2}$. Letting $t = \frac{nx}{\pi} > 0$, we rewrite the CDF of X:

$$F_X\left(\frac{nx}{\pi}\right) = \frac{1}{2} + \frac{1}{\pi}\left(\frac{\pi}{2} - \arctan\left(\frac{\pi}{nx}\right)\right) = 1 - \frac{1}{\pi}\arctan\left(\frac{\pi}{nx}\right).$$

Substituting this into (2) yields:

$$F_n(x) = \left(1 - \frac{1}{\pi}\arctan\left(\frac{\pi}{nx}\right)\right)^n$$
$$= \exp\left[n\ln\left(1 - \frac{1}{\pi}\arctan\left(\frac{\pi}{nx}\right)\right)\right].$$

To evaluate the limit of the exponent, let $z = \frac{1}{n}$. As $n \to \infty$, $z \to 0^+$:

$$\lim_{n\to\infty} n\ln\left(1 - \frac{1}{\pi}\arctan\left(\frac{\pi}{nx}\right)\right) = \lim_{z\to 0^+} \frac{\ln\left(1 - \frac{1}{\pi}\arctan\left(\frac{\pi z}{x}\right)\right)}{z}$$
$$\overset{\text{L'H}}{=} \lim_{z\to 0^+} \frac{\left(1 - \frac{1}{\pi}\arctan\left(\frac{\pi z}{x}\right)\right)^{-1} \cdot \left(-\frac{1}{\pi}\right) \cdot \left(1 + \left(\frac{\pi z}{x}\right)^2\right)^{-1} \cdot \frac{\pi}{x}}{1}$$
$$= \frac{1 \cdot \left(-\frac{1}{\pi}\right) \cdot 1 \cdot \frac{\pi}{x}}{1}$$
$$= -\frac{1}{x}.$$

By the continuity of the exponential function, we conclude $\lim_{n\to\infty} F_n(x) = \exp\left(-\frac{1}{x}\right)$ for $x > 0$.

Combining both cases, we have established that the sequence of distribution functions converges pointwise for all $x \in \mathbb{R}$ to the limit function $F_M(x)$:

$$\lim_{n\to\infty} F_n(x) = F_M(x) := \begin{cases} \exp(-1/x) & \text{if } x > 0, \\ 0 & \text{if } x \le 0. \end{cases}$$

The limit function $F_M(x)$ is a valid distribution function and is continuous everywhere on $\mathbb{R}$ (specifically at $x = 0$, since $\lim_{x\to 0^+} e^{-1/x} = 0$). By Theorem 4.1.6, pointwise convergence of the CDF at all continuity points of the limit CDF implies convergence in distribution. Therefore, $\frac{\pi M_n}{n} \xrightarrow{d} M$.

Part c)

We will prove this utilizing characteristic functions and Lévy's Continuity Theorem (Theorem 4.2.4). Since $X_n \sim \mathcal{N}(\mu_n, \sigma_n^2)$, we know from Example 3.5.3 (item 5) that its characteristic function is:

$$\varphi_{X_n}(u) = \exp\left(iu\mu_n - \frac{1}{2}\sigma_n^2 u^2\right), \quad u \in \mathbb{R}.$$

We are given that $X_n \xrightarrow{d} X$. By Lévy's Continuity Theorem (Theorem 4.2.4 a), the sequence of characteristic functions $\varphi_{X_n}(u)$ must converge pointwise to $\varphi_X(u)$, where φ_X is the characteristic function of X. Furthermore, $\varphi_X(u)$ must be continuous at $u = 0$ with $\varphi_X(0) = 1$.

Taking the absolute value of the characteristic functions, we get:

$$|\varphi_{X_n}(u)| = \exp\left(-\frac{1}{2}\sigma_n^2 u^2\right) \xrightarrow{n\to\infty} |\varphi_X(u)|.$$

Since φ_X is continuous at 0 and $\varphi_X(0) = 1$, there must exist a small neighborhood around 0 where φ_X is non-zero. Let $u_0 \neq 0$ be a point in this neighborhood such that $|\varphi_X(u_0)| > 0$. We then have:

$$\exp\left(-\frac{1}{2}\sigma_n^2 u_0^2\right) \xrightarrow{n\to\infty} |\varphi_X(u_0)| > 0.$$

Taking the natural logarithm (which is continuous on $(0, \infty)$):

$$-\frac{1}{2}\sigma_n^2 u_0^2 \xrightarrow{n\to\infty} \ln|\varphi_X(u_0)| \implies \sigma_n^2 \xrightarrow{n\to\infty} -\frac{2}{u_0^2}\ln|\varphi_X(u_0)|.$$

Let $\sigma^2 := -\frac{2}{u_0^2}\ln|\varphi_X(u_0)|$. Since $|\varphi_X(u_0)| \le 1$, the logarithm is non-positive, meaning $\sigma^2 \ge 0$. Thus, the variance sequence converges to a finite, non-negative limit: $\sigma_n^2 \to \sigma^2$.

Knowing that $\sigma_n^2 \to \sigma^2$, we can analyze the phase of the characteristic function. For any $u \in \mathbb{R}$:

$$\varphi_{X_n}(u) = \exp(iu\mu_n)\exp\left(-\frac{1}{2}\sigma_n^2 u^2\right) \xrightarrow{n\to\infty} \varphi_X(u).$$

Since $\exp(-\frac{1}{2}\sigma_n^2 u^2) \to \exp(-\frac{1}{2}\sigma^2 u^2)$ and this exponential is strictly greater than zero, we can isolate the phase component:

$$\exp(iu\mu_n) \xrightarrow{n\to\infty} \frac{\varphi_X(u)}{\exp\left(-\frac{1}{2}\sigma^2 u^2\right)} =: g(u). \tag{3}$$

We now show that the sequence of means $(\mu_n)_{n\in\mathbb{N}}$ must also converge to a finite limit. First, we rule out that $|\mu_n| \to \infty$. By integrating the left side of (3) over a symmetric interval $[-\delta, \delta]$, we obtain $\int_{-\delta}^{\delta} \exp(iu\mu_n)du = \frac{2\sin(\delta\mu_n)}{\mu_n}$. If $|\mu_n| \to \infty$, this integral would converge to 0. However, the integral of the right side, $\int_{-\delta}^{\delta} g(u)du$, does not equal zero for sufficiently small $\delta > 0$ because $g(0) = 1$ and g is continuous at 0. Thus, $(\mu_n)_{n\in\mathbb{N}}$ must be bounded. Let μ and μ' be two limit points of the bounded sequence $(\mu_n)_{n\in\mathbb{N}}$. Then, taking the limit along the corresponding subsequences in (3) gives $\exp(iu\mu) = g(u)$ and $\exp(iu\mu') = g(u)$ for all $u \in \mathbb{R}$. This implies $\exp(iu(\mu - \mu')) = 1$ for all $u \in \mathbb{R}$, which is only mathematically possible if $\mu = \mu'$. Because (μ_n) is bounded and has exactly one limit point, the sequence itself converges to $\mu \in \mathbb{R}$.

Finally, since $\mu_n \to \mu$ and $\sigma_n^2 \to \sigma^2$, the characteristic function of X_n converges as follows:

$$\lim_{n\to\infty} \varphi_{X_n}(u) = \lim_{n\to\infty} \exp\left(iu\mu_n - \frac{1}{2}\sigma_n^2 u^2\right) = \exp\left(iu\mu - \frac{1}{2}\sigma^2 u^2\right).$$

By Lévy's Continuity Theorem (Theorem 4.2.4), $\varphi_X(u) = \exp(iu\mu - \frac{1}{2}\sigma^2 u^2)$. This is exactly the characteristic function of a $\mathcal{N}(\mu, \sigma^2)$ distribution (if $\sigma^2 = 0$, it represents the degenerate Dirac measure δ_μ). By the Uniqueness Theorem (Theorem 3.5.4), $X \sim \mathcal{N}(\mu, \sigma^2)$.